

UTSAV TRIPATHI

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CAREER OBJECTIVE

Highly skilled and accomplished Senior Data Scientist with a Master's in Statistics and extensive experience in statistical modeling, machine learning, and Generative AI application development. Proven track record of building predictive frameworks, deploying innovative AI agents, and leveraging cutting-edge methodologies to drive strategic business outcomes across FinTech, Ad-Tech, and Insurance sectors.

WORK EXPERIENCE

EY (Ernst & Young) — Senior Data Scientist

Sep 2023 – Present

Project 1: Generative AI Agent Development & Innovation

- Collaborated across cross-functional teams to design, develop, and deploy autonomous AI agents tailored for complex enterprise workflows and client-facing automation.
- Utilized Vibe Coding techniques to accelerate rapid prototyping, iterative refinement, and functional enhancement of LLM-driven architectures.
- Conducted rigorous research on emerging trends in foundational models, prompt engineering, and RAG frameworks to recommend actionable tech-stack improvements.
- Implemented robust testing, prompt evaluation, and debugging protocols to optimize agent latency, accuracy, and overall performance.
- Led knowledge-sharing sessions and technical workshops to train internal staff and stakeholders on effectively interacting with and maximizing AI agent capabilities.

Project 2: Credit Risk Analysis

- Built end-to-end predictive models to analyze customer payment behavior and identify distinct patterns indicating potential installment defaults.
- Responsibilities included: Spearheaded base data creation, rigorous data preprocessing, variable reduction, statistical model development, validation, and comprehensive technical documentation.

Project 3: Product Propensity Modeling

- Led the retail lending team to strategize and execute 'Next Best Product' recommendations for Personal Loan (PL), Business Loan (BL), Mortgage Loan Against Property (MLAP), and Credit Card (CC) customers.
- Developed and tuned high-performing product-level propensity models using XGBoost and Random Forest.
- Opportunity Sizing: Extracted complex relational data from SQL Server databases using Python and SQL to identify and map out multi-channel product opportunities for RCMB customers.

Bajaj General Insurance — Deputy Manager Analytics

May 2022 – Aug 2023

- **Fraud Detection Modeling:** Developed a high-impact predictive model to flag anomalous patterns and identify fraudulent insurance claims early in the lifecycle.
- **Customer Segmentation:** Led a strategic segmentation project employing K-Means and Hierarchical Clustering algorithms to group customers by demographic, risk profiles, and historical claim behavior, significantly enhancing targeted fraud mitigation.
- **NLP & Sentiment Analysis:** Engineered an NLP pipeline to process customer reviews across policies and renewals. Cleaned text, extracted sentiment scores, and classified clients into Promoters (58%), Detractors (22%), and Neutrals (19%) to drive Net Promoter Score (NPS) insights across various lines of business.
- **Business Intelligence:** Built interactive Power BI dashboards leveraging AWS Athena (SQL) and Python to monitor critical KPIs, top fraudulent brokers, and high-risk claim locations.
- **Tools used:** Python, AWS Athena, SQL, SAS, Excel, Qlik Sense, Power BI.

IRICBS — Assistant Manager - Statistics

Mar

2019 – May 2022

- **Insurance Cross-Selling Prediction:** Built predictive models utilizing feature engineering and derived variables to forecast existing customer interest in secondary vehicle insurance products.
- **Credit Default Modeling:** Developed a risk framework to predict loan defaults, rigorously validated against Out-of-Time (OOT) data samples.
- **Agricultural Insurance Pricing:** Utilized advanced pricing methodologies to statistically estimate premium rates and loss costs based on crop cutting experiments and real-time rainfall report analysis.
- **Tools used:** Python, Amazon Athena, SQL, Qlik Sense.

Kotak Life Insurance — Intern

Aug 2018 – Jan 2019

- Estimated risk profiles for financial institution groups to support strategic product pricing.
- Designed and built automated calculation tools and pricing models using SQL and MS Excel.

TECHNICAL SKILLS

- **Generative AI:** Large Language Models (LLMs), AI Agent Frameworks, Vibe Coding, Prompt Engineering, RAG
- **Data Analysis Tools:** Python (NumPy, Pandas, Scikit-Learn), R, SAS, SPSS, Minitab, Google Colab
- **Database Management:** MySQL, Microsoft SQL Server, AWS Athena
- **Data Visualization:** Power BI, Qlik Sense, Tableau
- **Cloud & Big Data:** Amazon AWS, Microsoft Azure

ACADEMIC PROJECT

Project Name: “Farming through perspective of non-farmers”

Objective: Statistical study evaluating the economic perception and justification of agricultural loan waivers among regular taxpayers.

Tools Used: SAS, Google Sheets, Google Forms

EDUCATION

Course	University / Board	Year	Percentage / Grade
M.Sc. Statistics	University of Mumbai	2018	6.38 / 10 (SGPI)
B.Sc. (Statistics)	University of Mumbai	2016	83% (7/7 Grade)
H.S.C.	Maharashtra Board	2013	62.67%
S.S.C.	Maharashtra Board	2011	84.55%

CO-CURRICULAR ACTIVITIES

- Worked as a volunteer at STATISTIKA held on 11th Jan 2016 at Ramniranjan Jhunjhunwala College.

PERSONAL DETAILS

- **Languages Known:** English, Hindi & Marathi
- **Nationality:** Indian

Declaration: I hereby declare that all the details furnished above are true to the best of my knowledge and belief.

Place: Mumbai

Date: _____